

Smart Thermostat Alert System

Andy Zheng CNIT 549 Internet of Things Security

11/03/2025

IoT Devices

Physical IoT Devices:

- Raspberry Pi 4 Model B (2 GB) – Serves as the main controller
- HC-SR501 PIR Sensor – Detects Motion
- AHT21 Sensor – Measures Temperature and Humidity

Virtual/ Cloud Devices

- AWS IoT Core – used for the MQTT client
- AWS SNS – Used for the alert system
- AWS S3 - Used to store data for future analytical usage
- Amazon Athena – Used to query information from S3 buckets
- Amazon Quick Insights – Dashboard to present data

Communication Protocol

Sensors are connected locally on the Raspberry Pi 4 Model B (2 GB)

Data is published MQTT over TLS 1.3 to Topic

	Name	Status	Date updated	Security policy
☐	lot:Data-ATS	✔ Enabled	October 27, 2025, 18:00:55 (UTC-04:00)	IoTSecurityPolicy_TLS13_1_2_2022_10

IoT Architecture



Physical Layer

HC-SR501 and AHT21 collecting data



Edge layer

Raspberry Pi 4 processes sensor data locally and formats MQTT messages.



Cloud layer

AWS IoT Core receives messages which creates a message routing rule→ sends alerts via Amazon SNS.



Cloud Storage

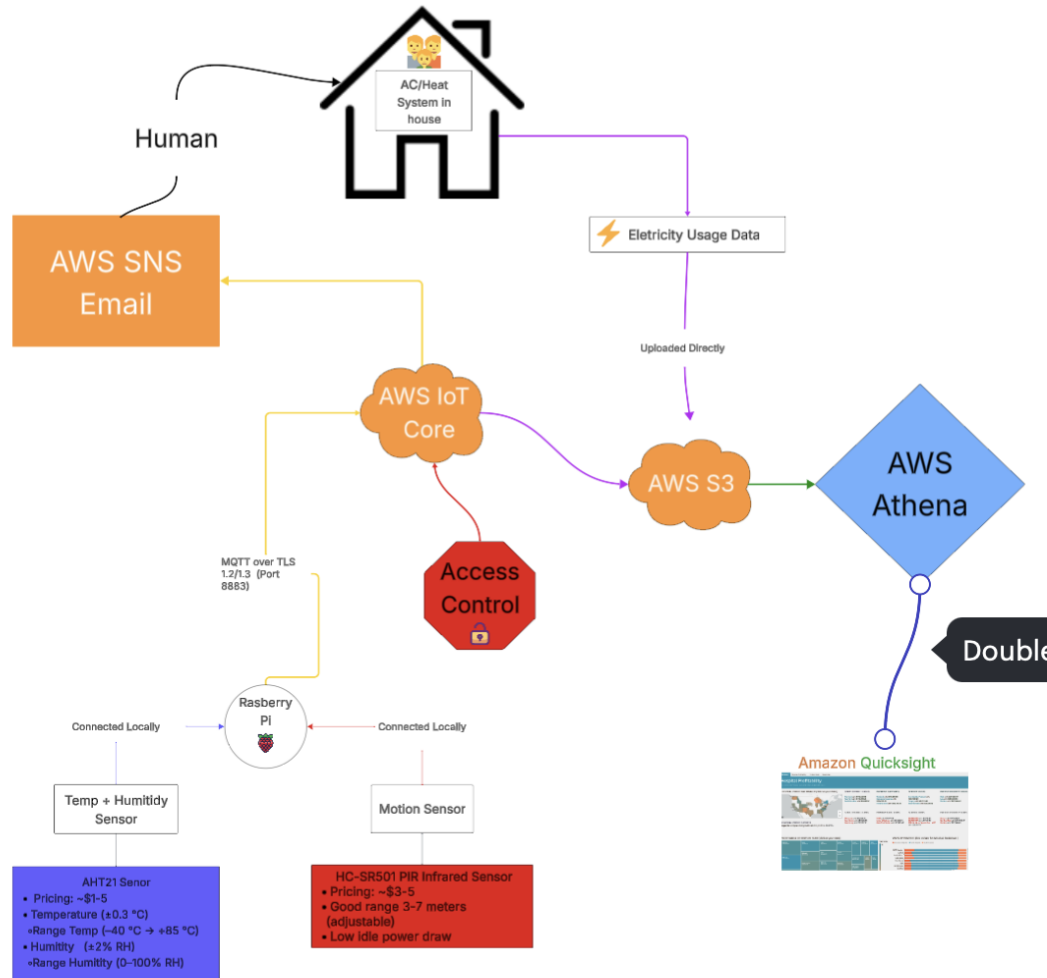
Data will be stored in S3 bucket using Messaging Routing Rules



Live Analytics

Data will be queried from AWS S3 bucket to Amazon Athena and then transferred to AWS QuickSight

Current Smart Thermostat Alert System Model

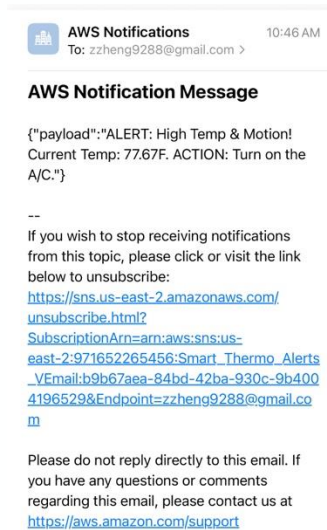


Smart Thermostat Alert Steps

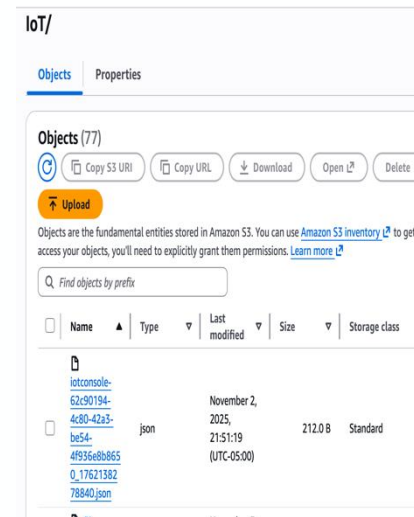
```
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Nov 3 16:02:22 2025 from 192.168.4.20
smartthermo@SmartTPI:~$ cd Documents
smartthermo@SmartTPI:~/Documents$ cd
smartthermo@SmartTPI:~$ source ~/venvs/ah2/bin/activate
(ah2) smartthermo@SmartTPI:~$ cd Documents
(ah2) smartthermo@SmartTPI:~/Documents$ python3 Smart_Thermo_FinalVersionFinal.py
PIR (GPIO23) + AHT21 Module Initialized, please wait 60 seconds to get started
Ready
Connecting to AWS IoT...
Connected to AWS IoT Core.
Time since last motion: N/A (first detection)
Motion detected!
Temp: 75.49 F Humidity: 50.83 %
[TX] smartthermo/Pi-Thermostat-v1/telemetry: {'ts': 1762451482, 'device': 'Pi-Thermostat-v1', 'motion': True, 'temperature_f': 75.49, 'humidity_pct': 50.83}
^CExiting program
Clean exit.
(ah2) smartthermo@SmartTPI:~/Documents$
```

Step 1: Collect data



Step 2: Get an Alert



Step 3: Data automatically goes to S3

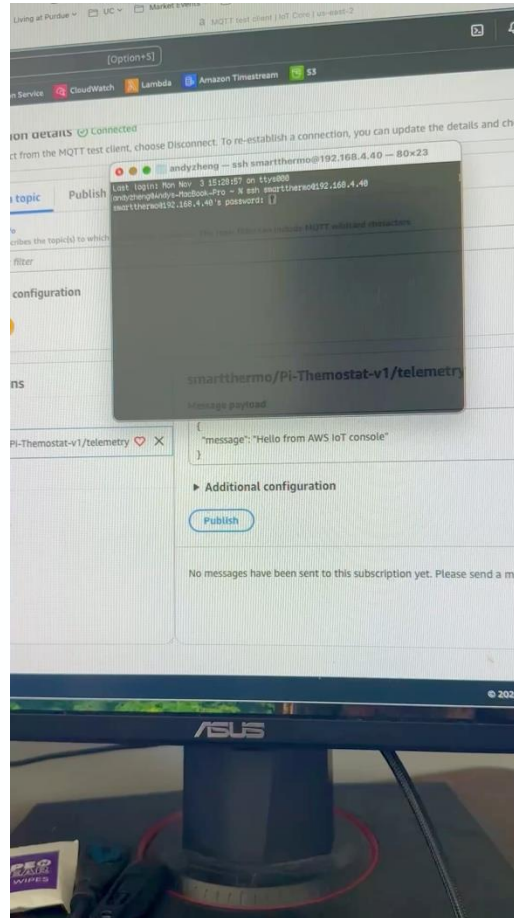


Step 4: Analyze and Present Data

Amazon QuickSight IoT Example



Smart Thermostat Alert Shown



Challenges and Troubleshooting

- PIR false motion triggers - adjusted the onboard potentiometers for sensitivity and delay; added debounce logic in code
- AHT21 came unsoldered – I2C device did not appear using i2cdetect, soldered equipment and retested
- MQTT connection errors: Solved by verifying IoT policy permissions and updating root CA certificates on the Pi.
- AWS SNS Text Message not available if there is not an origination entity, using AWS SNS Email Message instead
- IAM policy for S3 writing – created a customer S3:PutObject for the AWS, attached an IoT role and tested with simulated message
- AWS Timestream is not free and can't be used anymore, planning to use AWS Athena with Amazon Quick Sight

Security Analysis Plan

- "Recon & Inventory: Identify Pi IP, Open Ports, AWS endpoints (Thing, Bucket, Athena, Quick Insight)
- Network Capture: Observe traffic between raspberry Pi and AWS IoT Core using Wireshark
- MQTT Replay Test: Subscribe to telemetry topic, send a benign replayed message; observe if Bucket or Athena receives duplicated data
- Sensor Spoof Demo: Locally simulate false temperature/motion data to test logic integrity
- Filesystem Review: Inspect PI configs for stored keys, endpoint URLs, or credentials
- Cloud IAM Review: Examine IoT Roles, S3 Bucket Policy, and Athena/Quicksight permissions
- S3 Access Test: Verify that devices are private and encrypted, and only writeable by the IoT Thing role

References

- AWS QuickSight Solutions Gallery. (n.d.).
<https://d2lvqq4w5ulk4.cloudfront.net/?dashboardName=healthcare>
- Configuration and vulnerability analysis in AWS IOT sitewise - AWS IOT SiteWise. (n.d.).
<https://docs.aws.amazon.com/iot-sitewise/latest/userguide/vulnerability-analysis-and-management.html>
- Dinculeană, D., & Cheng, X. (2019, February 27). Vulnerabilities and limitations of MQTT protocol used between IOT devices. MDPI. <https://www.mdpi.com/2076-3417/9/5/848#:~:text=The%20broker%20receives%20published%20messages,to%20protect%20the%20transferred%20information.>
- HIBCTR aht21 temperature humidity sensor - fast reading system. Future of Education. (2025, October 22). https://therisk.global/education/product/hibctr-aht21-temperature-humidity-sensor-fast-reading-system/?srsId=AfmBOopE_HmReadHscoGs8PO-rYyIRvisALNAZPPIDJuLPGUIEoaOJC_
- Manik, V. K. (2023, September 1). A step-by-step guide to data management and analysis using AWS: S3, Athena, and quicksight. Medium.
<https://varunmanik1.medium.com/a-step-by-step-guide-to-data-management-and-analysis-using-aws-s3-athena-and-quicksight-1d1d49e766b6>
- Online. (n.d.). HC-SR501 PIR MOTION DETECTOR Product Discription.
- Thermostat IoT Device Monitoring. AWS QuickSight Solutions Gallery. (n.d.).
<https://d2lvqq4w5ulk4.cloudfront.net/?dashboardName=iot>